

程序一：（平面拟合）

```
m=数据一.xls;
a=m(:,1:3);
a1=m(:,4:6);
a2=m(:,7:9);
a3=m(:,10:12);
hold on;
c=[1,8;9,16;16,24;24,32;32,40;40,48;48,56;56,64;64,72;72,80;80,88;88,96;96,104;];
b=zeros(3,13);
for k=1:13
x=a([c(k,1):c(k,2)],1);
y=a([c(k,1):c(k,2)],2);
z=a([c(k,1):c(k,2)],3);
Xcolv = x(:);
Ycolv = y(:);
Zcolv = z(:);
Const = ones(size(Xcolv));
Coefficients1(:,k) = [Xcolv Ycolv Const]\Zcolv;
XCoeff = Coefficients1(1);
YCoeff = Coefficients1(2);
CCoeff = Coefficients1(3);
L=plot3(x,y,z,'ro'); % 绘制三维图形
set(L,'Markersize',2*get(L,'Markersize'))
set(L,'Markerfacecolor','r')
hold on;
grid on;
[xx, yy]=meshgrid(0:0.1:1,0:0.1:1);
zz = XCoeff * xx + YCoeff * yy + CCoeff;
surf(xx,yy,zz)
title(sprintf('Plotting plane z=(%f)*x+(%f)*y+(%f)',XCoeff,
YCoeff, CCoeff))
end
```

Coefficients1%由程序计算所得的系数

```
for k=1:13
x=a1([c(k,1):c(k,2)],1);
y=a1([c(k,1):c(k,2)],2);
z=a1([c(k,1):c(k,2)],3);
Xcolv = x(:);
Ycolv = y(:);
Zcolv = z(:);
Const = ones(size(Xcolv));
Coefficients2(:,k) = [Xcolv Ycolv Const]\Zcolv;
XCoeff = Coefficients2(1);
YCoeff = Coefficients2(2);
CCoeff = Coefficients2(3);
```

```

L=plot3(x,y,z,'ro'); % 绘制三维图形
set(L,'Markersize',2*get(L,'Markersize'))
set(L,'Markerfacecolor','r')
hold on;
grid on;
[xx, yy]=meshgrid(0:0.1:1,0:0.1:1);
zz = XCoeff * xx + YCoeff * yy + CCoeff;
surf(xx,yy,zz)
title(sprintf('Plotting plane z=(%f)*x+(%f)*y+(%f)',XCoeff,
YCoeff, CCoeff))
end

```

Coefficients2%由程序计算所得的系数

```

for k=1:13
x=a2([c(k,1):c(k,2)],1);
y=a2([c(k,1):c(k,2)],2);
z=a2([c(k,1):c(k,2)],3);
Xcolv = x(:);
Ycolv = y(:);
Zcolv = z(:);
Const = ones(size(Xcolv));
Coefficients3(:,k) = [Xcolv Ycolv Const]\Zcolv;
XCoeff = Coefficients3(1);
YCoeff = Coefficients3(2);
CCoeff = Coefficients3(3);
L=plot3(x,y,z,'ro'); % 绘制三维图形
set(L,'Markersize',2*get(L,'Markersize'))
set(L,'Markerfacecolor','r')
hold on;
grid on;
[xx, yy]=meshgrid(0:0.1:1,0:0.1:1);
zz = XCoeff * xx + YCoeff * yy + CCoeff;
surf(xx,yy,zz)
title(sprintf('Plotting plane z=(%f)*x+(%f)*y+(%f)',XCoeff,
YCoeff, CCoeff))
end

```

Coefficients3%由程序计算所得的系数

m=[数据一.xls](#);

a=m(:,1:3);%1989 年观测数据

a1=m(:,4:6);%1999 年观测数据

a2=m(:,7:9);%2009 年观测数据

a3=m(:,10:12);%2011 年观测数据

for k=1:13

x=a3([c(k,1):c(k,2)],1);

y=a3([c(k,1):c(k,2)],2);

z=a3([c(k,1):c(k,2)],3);

Xcolv = x(:);

```

Ycolv = y(:);
Zcolv = z(:);
Const = ones(size(Xcolv));
Coefficients4(:,k) = [Xcolv Ycolv Const]\Zcolv;
XCoeff = Coefficients4(1);
YCoeff = Coefficients4(2);
CCoeff = Coefficients4(3);
L=plot3(x,y,z,'ro'); % 绘制三维图形
set(L,'Markersize',2*get(L,'Markersize'))
set(L,'Markerfacecolor','r')
hold on;
grid on;
[xx, yy]=meshgrid(0:0.1:1,0:0.1:1);
zz = XCoeff * xx + YCoeff * yy + CCoeff;
surf(xx,yy,zz)
title(sprintf('Plotting plane z=(%f)*x+(%f)*y+(%f)',XCoeff,
YCoeff, CCoeff))
end
Coefficients4%由程序计算所得的系数

```

程序二：（中心点坐标的确定）

首先应用matlab建立ff2.m文件：

```

function f=ff2(x,A,B)
x1=A(:,1);y=A(:,2);z=A(:,3);
f=sum((x1-x(1)).^2+(y-x(2)).^2+(z-(B(1)*x(1)+B(2)*x(2)+B(3))).^2);

```

然后，再运行如下程序：

```

m=数据一.xls;
F1=m(:,1:3);%1989 年观测数据
F2=m(:,4:6);%1999 年观测数据
F3=m(:,7:9);%2009 年观测数据
F4=m(:,10:12);%2011 年观测数据

```

```

P1=[-0.000830978 0.003417388 0.471956284
-0.000817778 0.003628518 5.887189422
-0.103531579 -0.170598336 160.1073406
-0.084434924 -0.139125499 137.2555526
-0.093172538 -0.156337182 155.8157108
-0.10905983 -0.155763638 169.0503192
-0.097249562 -0.132905095 154.0980362
-0.100944809 -0.138739994 162.7653381
-0.107075519 -0.14903722 175.1283597
-0.107132219 -0.156237569 182.2592083
-0.135031497 -0.217484619 234.2576489
-0.145292163 -0.233311605 252.6149174
-0.147842995 -0.253950265 268.9965036
];
P2=[-0.001487998 0.003714862 0.684399642
-0.000492827 0.003781979 5.617202869

```

```

-0.103877083 -0.170596425 160.2976036
-0.082898692 -0.139164862 136.4014982
-0.093547416 -0.156341154 156.0256571
-0.108695055 -0.155620482 168.7628486
-0.096893482 -0.132785851 153.8288996
-0.101347724 -0.138656209 162.9445585
-0.106635181 -0.148858581 174.7784003
-0.107683182 -0.156283925 182.5910487
-0.135533438 -0.21745348 234.5201978
-0.14590835 -0.233379092 252.9945534
-0.148170132 -0.254010047 269.2086747
];
P3=[-0.002373299 -0.003769787 5.079985167
-0.003971777 -0.000193548 9.66122192
0.171779992 -0.104986237 -30.24613939
0.140823091 -0.082759129 -19.89282395
0.157627698 -0.09089535 -20.55834552
0.168465935 -0.118761803 -7.632855986
0.134087929 -0.095297204 3.269995936
0.139156581 -0.100754259 6.766714304
0.147528599 -0.104817538 7.636402368
0.15272301 -0.103455455 7.301054717
0.215152629 -0.135879385 -6.995143348
0.232413579 -0.147310618 -6.561167274
0.2413723 -0.157728579 -2.08825351
];
P4=[-0.002333174 -0.003740287 5.040576042
-0.002554701 -0.002530475 10.06101404
0.172474839 -0.106260378 -29.98224837
0.14091257 -0.085030797 -18.77216725
0.158721438 -0.09191605 -20.65340477
0.168403679 -0.119171259 -7.390169205
0.133926192 -0.094930062 3.161969153
0.139361927 -0.100794271 6.667407016
0.148239426 -0.107208857 8.463752307
0.155180017 -0.107129134 7.808521389
0.216632193 -0.138845389 -6.295107563
0.231243928 -0.142257642 -8.553926047
0.241592539 -0.158407287 -1.864415599
];
c=[1,8;9,16;17,24;25,32;33,40;41,48;49,56;57,64;65,72;73,80;81,88;89,96;97,104;];
b1=zeros(3,13);
for k=1:13
A = F1([c(k,1):c(k,2)],:);%1986 年观测数据
a=P1(k,:);
b1(1:2,k) = fminsearch(@(x)ff2(x,A,a),[1,2]);
b1(3,k)=b1(1,k)*P1(k,1)+b1(2,k)*P1(k,2)+P1(k,3);
end
b2=zeros(3,13);
for k=1:13

```

```

A = F2([c(k,1):c(k,2)],:);%1996 年观测数据
a=P2(k,:);
b2(1:2,k) = fminsearch(@(x)ff2(x,A,a),[1,2]);
b2(3,k)=b2(1,k)*P2(k,1)+b1(2,k)*P2(k,2)+P2(k,3);
end
b3=zeros(3,13);
for k=1:13
A = F3([c(k,1):c(k,2)],:);%2009 年观测数据
a=P3(k,:);
b3(1:2,k) = fminsearch(@(x)ff2(x,A,a),[1,2]);
b3(3,k)=b3(1,k)*P3(k,1)+b3(2,k)*P3(k,2)+P3(k,3);
end
b4=zeros(3,13);
for k=1:13
A = F4([c(k,1):c(k,2)],:);%2011 年观测数据
a=P4(k,:);
b4(1:2,k) = fminsearch(@(x)ff2(x,A,a),[1,2]);
b4(3,k)=b4(1,k)*P4(k,1)+b4(2,k)*P4(k,2)+P4(k,3);
end
b1 ,b2,b3,b4

```

程序三：（倾斜角的计算）

```

clear
clc
a=[566.664741    567.2640516
522.7105282  522.2540994
1.787375037  55.10854517
566.6649707  567.2543
522.7101805  522.2366
1.783000777  55.11965447
566.7268041  567.336
522.7014735  522.2148
1.76449979   55.091
566.7269181  567.3375
522.7013548  522.2135
1.763250449  55.087
];%各次底层中心点及塔尖中心点的坐标值
b=zeros(4,1);
for i=1:4
b(i,1)=atan(sqrt((a(3*i-2,1)-a(3*i-2,2))^2+(a(3*i-1,1)-a(3*i-1,2))^2)/(a(3*i,2)-a(3*i,1)));
end
b

```

程序四：（空间曲线方程的拟合）

```

clear
clc
a1=[566.664741    566.7196032  566.7251066  566.7841861  566.824563   566.8660307
566.9166678  566.9538279  566.9897481  567.0267344  567.0564911  567.1007311
567.148014   567.2640516
522.7105282  522.668344   522.5474854  522.5418216  522.4961758  522.4639359

```

```

522.4669521 522.4505947 522.4317649 522.4184393 522.3456014 522.3016699
522.2614503 522.2540994
1.787375037 7.320250093 12.28766395 16.70028124 21.31764794 25.84712287
29.52711866 33.04949985 36.55586028 39.89098906 44.08502791 48.36058464
52.5192089 55.10854517
];
x=a1(1,:);
y=a1(2,:);
z=a1(3,:);
x1=polyfit(z,x,2)%z 为自变量, x 为因变量
y1=polyfit(z,y,2)%z 为自变量, y 为因变量
a2=[566.6649707 566.7205451 566.7265463 566.7869552 566.827267 566.8696053
566.9205503 566.9579355 566.9946391 567.0317229 567.0620474 567.1064981
567.1540086 567.2543
522.7101805 522.6674562 522.5459168 522.5394603 522.4932815 522.4605572
522.4630366 522.4463037 522.4268193 522.4132488 522.3398815 522.2955148
522.2552224 522.2366
1.783000777 7.314628384 12.2829702 16.69614048 21.31277587 25.84083613
29.52177461 33.04364328 36.54837315 39.88566431 44.07846016 48.35469052
52.51373486 55.11965447
];
x=a2(1,:);
y=a2(2,:);
z=a2(3,:);
x2=polyfit(z,x,2)%z 为自变量, x 为因变量
y2=polyfit(z,y,2)%z 为自变量, y 为因变量
a3=[566.7268041 566.763964 566.8798373 566.8829498 566.9237686 567.0100817
567.0214421 567.0722482 567.1256312 567.1797278 567.2555827 567.3031922
567.3514709 567.336
522.7014735 522.6692942 522.5896155 522.5816831 522.5500444 522.4897794 522.482311
522.449441 522.4153258 522.3647064 522.3068783 522.2648641 522.2190398
522.2148
1.76449979 7.309000247 12.2677573 16.68898035 21.30717388 25.83719934
29.50962342 33.03954325 36.54536393 39.88097158 44.08064918 48.35263811
52.48580886 55.091
];
x=a3(1,:);
y=a3(2,:);
z=a3(3,:);
x3=polyfit(z,x,2)%z 为自变量, x 为因变量
y3=polyfit(z,y,2)%z 为自变量, y 为因变量
a4=[566.7269181 566.7641741 566.8809476 566.8830262 566.9252064 567.0107413
567.0222307 567.0732454 567.1263284 567.1816335 567.257516 567.3043501
567.3528621 567.3375
522.7013548 522.6689947 522.5891489 522.5805209 522.5487839 522.4889408
522.4816388 522.44856 522.4139989 522.3625187 522.3044631 522.2651696
522.2174357 522.2135
1.763250449 7.290500213 12.2599313 16.67333868 21.29915907 25.83086076
29.50188293 33.03600551 36.52682599 39.86353266 44.07156577 48.33554872
52.48075565 55.087

```

```

];
x=a4(1,:);
y=a4(2,:);
z=a4(3,:);
x4=polyfit(z,x,2)%z 为自变量, x 为因变量
y4=polyfit(z,y,2)%z 为自变量, y 为因变量

```

程序五：（曲率 K 值的计算）

```

clear
clc
syms x1 y1 x2 y2 x3 y3 x4 y4 z;
x1=-0.00007*z^2+0.0064*z+566.65624;
y1=-0.00001*z^2-0.00863*z+522.71088;
x2=-0.00006*z^2+0.0069*z+566.65375;
y2=0.00001*z^2+0.00856*z+522.70036;
x3=0.00002*z^2+0.01078*z+566.70374;
y3=-0.00004*z^2-0.00682*z+522.70905;
x4=0.00002*z^2-0.01083*z+566.70353;
y4=-0.00002*z^2+0.01083*z+566.70353;
dx1=diff(x1,z,1)
ddx1=diff(x1,z,2)
dy1=diff(y1,z,1)
ddy1=diff(y1,z,2)
dx2=diff(x2,z,1)
ddx2=diff(x2,z,2)
dy2=diff(y2,z,1)
ddy2=diff(y2,z,2)
dx3=diff(x3,z,1)
ddx3=diff(x3,z,2)
dy3=diff(y3,z,1)
ddy3=diff(y3,z,2)
dx4=diff(x4,z,1)
ddx4=diff(x4,z,2)
dy4=diff(y4,z,1)
ddy4=diff(y4,z,2)
n=[1.787375037   7.320250093  12.28766395  16.70028124  21.31764794  25.84712287
    29.52711866  33.04949985  36.55586028  39.89098906  44.08502791  48.36058464
    52.5192089   55.10854517
    1.783000777  7.314628384  12.2829702   16.69614048  21.31277587  25.84083613
    29.52177461  33.04364328  36.54837315  39.88566431  44.07846016  48.35469052
    52.51373486  55.11965447
    1.76449979   7.309000247  12.2677573   16.68898035  21.30717388  25.83719934
    29.50962342  33.03954325  36.54536393  39.88097158  44.08064918  48.35263811
    52.48580886  55.091
    1.763250449  7.290500213  12.2599313   16.67333868  21.29915907  25.83086076
    29.50188293  33.03600551  36.52682599  39.86353266  44.07156577  48.33554872
    52.48075565  55.087
];
%中心点的 z 轴数据
for k=1:length(n)

```

```
a(k,1)=sqrt(((4/625-(7*n(1,k))/50000)*(-1/50000)-(-863/100000-n(1,k)/50000)*(-7/50000))^2+(-7/50000)^2+(-1/50000)^2)/sqrt(((4/625-(7*n(1,k))/50000)^2+(-863/100000-n(1,k)/50000)^2+1)^3);
```

```
b(k,1)=sqrt(((69/10000-(3*n(2,k))/25000)*(1/50000)-(107/12500-n(2,k)/50000)*(-3/25000))^2+(-3/25000)^2+(1/50000)^2)/sqrt(((69/10000-(3*n(2,k))/25000)^2+(107/12500-n(2,k)/50000)^2+1)^3);
```

```
c(k,1)=sqrt((((n(3,k))/25000+539/50000)*(-1/12500)-(-341/50000-n(3,k)/12500)*(7/50000))^2+(1/12500)^2+(-1/25000)^2)/sqrt((((n(3,k))/25000+539/50000)^2+(-341/50000-n(3,k)/12500)^2+1)^3);
```

```
d(k,1)=sqrt(((n(4,k)/25000-1083/100000)*(-1/25000)-(1083/100000-n(4,k)/25000)*(-1/25000))^2+(-1/25000)^2+(-1/25000)^2)/sqrt((((n(4,k)/25000-1083/100000)^2+(1083/100000-n(4,k)/25000)^2+1)^3);
```

```
end
```

a,b,c,d%各次测量的的曲率值

程序六：（扭转量的计算）

F=[数据二.xls](#);

```
P=[566.664741 522.7105282 566.6649707 522.7101805 566.7268041
522.7014735 566.7269181 522.7013548
566.7196032 522.668344 566.7205451 522.6674562 566.763964 522.6692942
566.7641741 522.6689947
566.7251066 522.5474854 566.7265463 522.5459168 566.8798373 522.5896155
566.8809476 522.5891489
566.7841861 522.5418216 566.7869552 522.5394603 566.8829498 522.5816831
566.8830262 522.5805209
566.824563 522.4961758 566.827267 522.4932815 566.9237686 522.5500444
566.9252064 522.5487839
566.8660307 522.4639359 566.8696053 522.4605572 567.0100817 522.4897794
567.0107413 522.4889408
566.9166678 522.4669521 566.9205503 522.4630366 567.0214421 522.482311
567.0222307 522.4816388
566.9538279 522.4505947 566.9579355 522.4463037 567.0722482 522.449441
567.0732454 522.44856
566.9897481 522.4317649 566.9946391 522.4268193 567.1256312 522.4153258
567.1263284 522.4139989
567.0267344 522.4184393 567.0317229 522.4132488 567.1797278 522.3647064
567.1816335 522.3625187
567.0564911 522.3456014 567.0620474 522.3398815 567.2555827 522.3068783
567.257516 522.3044631
567.1007311 522.3016699 567.1064981 522.2955148 567.3031922 522.2648641
567.3043501 522.2651696
567.148014 522.2614503 567.1540086 522.2552224 567.3514709 522.2190398
567.3528621 522.2174357
```

```
];
```

```
A1=F(:, [1:4]);
```

```
A2=F(:, [3:6]);
```

```
A3=F(:, [5:8]);
```

```
B1(:,1)=P(:,3)-P(:,1);
```

```
B1(:,2)=P(:,4)-P(:,2);
```

```
B2(:,1)=P(:,5)-P(:,3);
```

```
B2(:,2)=P(:,6)-P(:,4);
```

```
B3(:,1)=P(:,7)-P(:,5);
```

```
B3(:,2)=P(:,8)-P(:,6);
```

```

for m=1:8
x(m,1)=fminbnd(@(x)niuqv1(x,A1(m,:),B1(1,:)), -0.005,0.005);
end
for m=9:16
x(m,2)=fminbnd(@(x)niuqv1(x,A1(m,:),B1(2,:)), -0.005,0.005);
end
for m=17:24
x(m,3)=fminbnd(@(x)niuqv1(x,A1(m,:),B1(3,:)), -0.005,0.005);
end
for m=25:32
x(m,4)=fminbnd(@(x)niuqv1(x,A1(m,:),B1(4,:)), -0.005,0.005);
end
for m=33:40
x(m,5)=fminbnd(@(x)niuqv1(x,A1(m,:),B1(5,:)), -0.005,0.005);
end
for m=41:48
x(m,6)=fminbnd(@(x)niuqv1(x,A1(m,:),B1(6,:)), -0.005,0.005);
end
for m=49:56
x(m,7)=fminbnd(@(x)niuqv1(x,A1(m,:),B1(7,:)), -0.005,0.005);
end
for m=57:64
x(m,8)=fminbnd(@(x)niuqv1(x,A1(m,:),B1(8,:)), -0.005,0.005);
end
for m=65:72
x(m,9)=fminbnd(@(x)niuqv1(x,A1(m,:),B1(9,:)), -0.005,0.005);
end
for m=73:80
x(m,10)=fminbnd(@(x)niuqv1(x,A1(m,:),B1(10,:)), -0.005,0.005);
end
for m=81:88
x(m,11)=fminbnd(@(x)niuqv1(x,A1(m,:),B1(11,:)), -0.005,0.005);
end
for m=89:96
x(m,12)=fminbnd(@(x)niuqv1(x,A1(m,:),B1(12,:)), -0.005,0.005);
end
for m=97:104
x(m,13)=fminbnd(@(x)niuqv1(x,A1(m,:),B1(13,:)), -0.005,0.005);
end
a1(:,1)=asin(sum(x)/8);
for m=1:8
x(m,1)=fminbnd(@(x)niuqv1(x,A2(m,:),B2(1,:)), -0.005,0.005);
end
for m=9:16
x(m,2)=fminbnd(@(x)niuqv1(x,A2(m,:),B2(2,:)), -0.005,0.005);
end
for m=17:24
x(m,3)=fminbnd(@(x)niuqv1(x,A2(m,:),B2(3,:)), -0.005,0.005);
end
for m=25:32
x(m,4)=fminbnd(@(x)niuqv1(x,A2(m,:),B2(4,:)), -0.005,0.005);
end
for m=33:40
x(m,5)=fminbnd(@(x)niuqv1(x,A2(m,:),B2(5,:)), -0.005,0.005);
end
for m=41:48
x(m,6)=fminbnd(@(x)niuqv1(x,A2(m,:),B2(6,:)), -0.005,0.005);

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end
for m=49:56
x(m,7)=fminbnd(@(x)niuqv1(x,A2(m,:),B2(7,:)), -0.005,0.005);
end
for m=57:64
x(m,8)=fminbnd(@(x)niuqv1(x,A2(m,:),B2(8,:)), -0.005,0.005);
end
for m=65:72
x(m,9)=fminbnd(@(x)niuqv1(x,A2(m,:),B2(9,:)), -0.005,0.005);
end
for m=73:80
x(m,10)=fminbnd(@(x)niuqv1(x,A2(m,:),B2(10,:)), -0.005,0.005);
end
for m=81:88
x(m,11)=fminbnd(@(x)niuqv1(x,A2(m,:),B2(11,:)), -0.005,0.005);
end
for m=89:96
x(m,12)=fminbnd(@(x)niuqv1(x,A2(m,:),B2(12,:)), -0.005,0.005);
end
for m=97:104
x(m,13)=fminbnd(@(x)niuqv1(x,A2(m,:),B2(13,:)), -0.005,0.005);
end
a1(:,2)=asin(sum(x)/8);
for m=1:8
x(m,1)=fminbnd(@(x)niuqv1(x,A3(m,:),B3(1,:)), -0.005,0.005);
end
for m=9:16
x(m,2)=fminbnd(@(x)niuqv1(x,A3(m,:),B3(2,:)), -0.005,0.005);
end
for m=17:24
x(m,3)=fminbnd(@(x)niuqv1(x,A3(m,:),B3(3,:)), -0.005,0.005);
end
for m=25:32
x(m,4)=fminbnd(@(x)niuqv1(x,A3(m,:),B3(4,:)), -0.005,0.005);
end
for m=33:40
x(m,5)=fminbnd(@(x)niuqv1(x,A3(m,:),B3(5,:)), -0.005,0.005);
end
for m=41:48
x(m,6)=fminbnd(@(x)niuqv1(x,A3(m,:),B3(6,:)), -0.005,0.005);
end
for m=49:56
x(m,7)=fminbnd(@(x)niuqv1(x,A3(m,:),B3(7,:)), -0.005,0.005);
end
for m=57:64
x(m,8)=fminbnd(@(x)niuqv1(x,A3(m,:),B3(8,:)), -0.005,0.005);
end
for m=65:72
x(m,9)=fminbnd(@(x)niuqv1(x,A3(m,:),B3(9,:)), -0.005,0.005);
end
for m=73:80
x(m,10)=fminbnd(@(x)niuqv1(x,A3(m,:),B3(10,:)), -0.005,0.005);
end
for m=81:88
x(m,11)=fminbnd(@(x)niuqv1(x,A3(m,:),B3(11,:)), -0.005,0.005);
end
for m=89:96

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x(m,12)=fminbnd(@(x)niuqvl(x,A3(m,:),B3(12,:)), -0.005,0.005);
end
for m=97:104
x(m,13)=fminbnd(@(x)niuqvl(x,A3(m,:),B3(13,:)), -0.005,0.005);
end
a1(:,3)=asin(sum(x)/8);

```

程序七：（灰色模型求值）

```

clc,clear
x0=[0.0141,0.0142,0.0146,0.0147];%原始数据序列
n=length(x0);
a_x0=diff(x0);%求 1 次累减序列，即 1 阶向前差分
B=[-x0(2:end)',ones(n-1,1)];
u=B\ a_x0%最小二乘拟合参数
x=dsolve('D2x+a*Dx=b','x(0)=c1,Dx(0)=c2');%求二阶微分方程的符号解
x=subs(x',{'a','b','c1','c2'},{u(1),u(2),x0(1),x0(1)});
yuce=subs(x,'t',0:n-1)%求已知数据点 1 次累加序列的预测值
x=vpa(x,6)
x0_hat=[yuce(1),diff(yuce)]%求已知数据点的预测值
epsilon=x0-x0_hat%求残差
delta=abs(epsilon./x0)%求相对误差

```